MATEUS CARNIEL BRAMBILLA
Software Engineer | Backend | Python & TypeScript
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PROFESSIONAL PROFILE

Backend-focused Software Engineer with professional experience in Python, Flask, Pydantic, and data/IoT system integration. Built and deployed a containerized RFID ingestion service, internal automation tooling with the Databricks SDK, and LLM integrations. Through projects, develops Node.js/TypeScript APIs with Express, PostgreSQL, Prisma, Zod, JWT/OAuth2, and API security practices.

ACADEMIC BACKGROUND

- **Bachelor of Science in Software Engineering — Univates** | 2025 – Expected Completion: 12/2028

CERTIFICATIONS AND COMPLEMENTARY TRAINING

- Google Cloud - Associate Cloud Engineer;
- Astronomer — Apache Airflow 3 Fundamentals (2026);
- Cisco Networking Academy — CCNA 1 & CCNA 2;
- Linux Fundamentals (Red Hat Training);
- Advanced SQL (HackerRank).

PROFESSIONAL EXPERIENCE

Tramontina Multi S.A. — Developer (Backend & Data Engineering) | Nov/2025 – Present

- **Backend & IoT:** Built and deployed a Flask/Pydantic service that receives RFID readings from a Zebra FX9600, applies business rules, and performs upserts into MongoDB and SQL Server; the application runs in Docker on a VM.
- **Platform Tooling:** Built an internal Python library with the Databricks SDK to export, parameterize, version, and promote Genie Space metadata across workspaces, enabling Git-based management.
- **AI Integration:** Optimized an LLM-based cataloging pipeline, reducing output cost by 75%, and implemented a primary/fallback model strategy through Databricks AI Gateway with exponential backoff and jitter.
- **Backend Automation:** Built a Python automation that collects web and PDF menu content, structures the data, uses an LLM for recommendations, and sends the result to the team through a webhook.

Tramontina Multi S.A. — Industrial Processes Trainee | Apr/2025 – Nov/2025

- **Technical Foundations:** Deepened Python, PySpark, Databricks, data analysis, Git, and software development fundamentals during the transition into the current role.

FEATURED PROJECTS

- **CronGoal — Backend API (Personal/Academic Project):** Node.js/TypeScript and Express 5 API for a productivity application, using PostgreSQL/Prisma with separation across routes, controllers, and services.
    - **Backend Engineering:** Implemented local/JWT authentication and Google OAuth2 with Passport.js, Zod validation, RBAC, Prisma ORM, object storage, and Helmet, CORS, and Rate Limiting middleware.
- **OLX Scraper (in development):** Python/FastAPI service for collecting and structurally processing listings, used as a hands-on backend and automation project.
- **Agent Orchestrator (PoC):** Built a supervisor for dynamically routing requests across specialized AI services in the Databricks ecosystem.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, Flask, FastAPI, TypeScript, Node.js, Express.js.
- **Backend & APIs:** Pydantic, Zod, Prisma, SQLAlchemy, Passport.js, REST APIs, Webhooks.
- **DevOps & Infrastructure:** Docker, Docker Compose, Git, GitHub Actions (fundamentals), Apache Airflow 3, Linux, VMs; Google Cloud Associate Cloud Engineer certified.
- **Data & Security:** PostgreSQL, SQL Server, MongoDB, JWT/OAuth2, Helmet, CORS, Rate Limiting.
- **Engineering Practices:** Clean Architecture, SOLID, Code Review, layered architecture, structured logging.

ACHIEVEMENTS

- **Global Nominee — NASA Space Apps Challenge (2025):** Winner of the local stage (Lajeado) and global nominee for the AirMood project, which I developed end-to-end while leading the team.